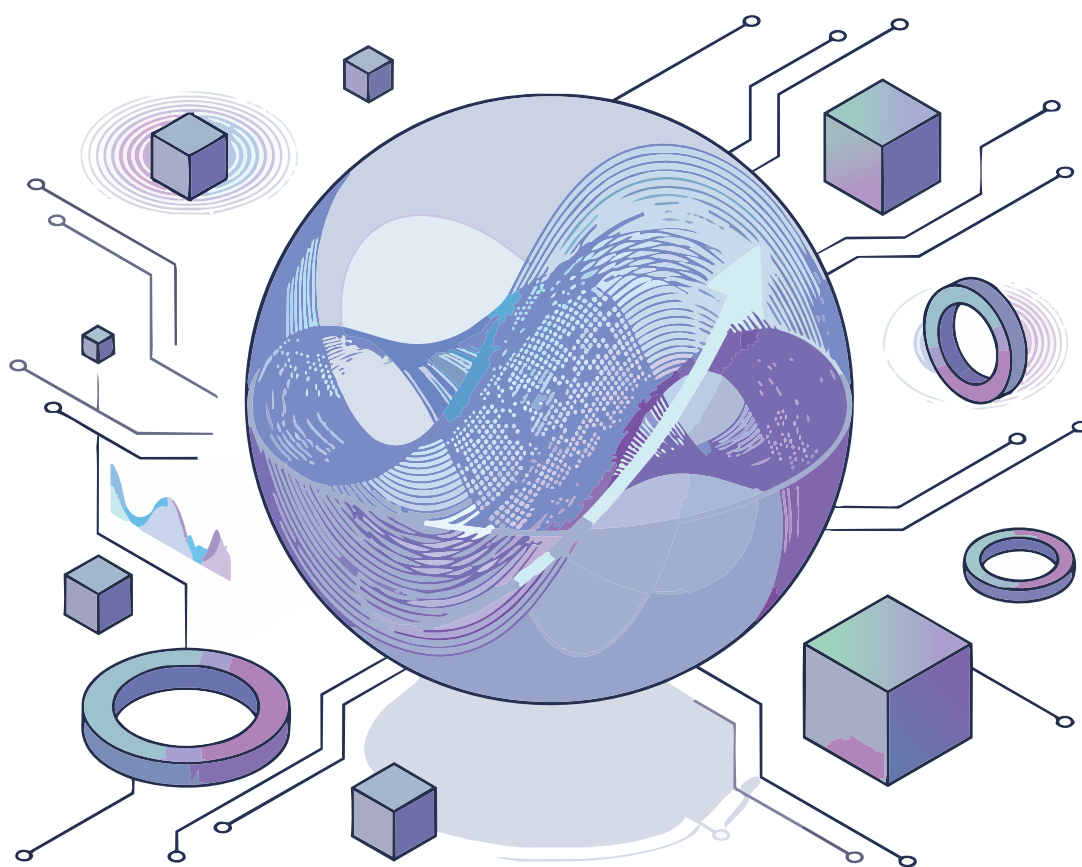


# Quantum Computing: The Strange World of Phase

In this third article in the series on quantum computing, we will develop an intuitive understanding of the quantum phase and explore how it influences the behaviour of quantum systems.



In the second article in this series on quantum computing (*OSFY*, July 2026), we explored several competing hardware architectures for building quantum computers and examined how quantum measurement enables us to extract classical information from a quantum system. We concluded by studying the Pauli-X gate, perhaps the most intuitive of the

three Pauli gates (Pauli-X, Pauli-Y, and Pauli-Z), owing to its close resemblance to the classical NOT gate. However, as we noted at the end of that article, the remaining two Pauli gates cannot be fully understood without first introducing one of the most fundamental—and often one of the most challenging—concepts in quantum computing: quantum phase.

By now, I assume that readers following this series have become familiar with several fundamental concepts of quantum computing, including qubits, superposition, quantum measurement, the Hadamard and Pauli-X gates, the Bloch sphere, the matrix representation of quantum states and gates, basic programming with Qiskit, and a brief overview of